

Part 1: HTML

1. Semantic Tags

Q1. What are semantic HTML tags?

Ans: Semantic tags clearly describe their meaning to both the browser and developers. Examples include `<header>`, `<footer>`, `<article>`, and `<section>`.

Q2. Why should we use semantic tags?

Ans: They improve accessibility, SEO, code readability, and make the document structure more meaningful.

Q3. What is the difference between `<div>` and `<section>`?

Ans: `<div>` is a non-semantic container, while `<section>` represents a meaningful section of content.

Q4. Name some commonly used semantic tags.

Ans: `<header>`, `<nav>`, `<main>`, `<article>`, `<section>`, `<footer>`, `<aside>`.

2. Attributes

Q5. What are HTML attributes?

Ans: Attributes provide additional information about an element and are written inside the opening tag.

Q6. What is the difference between global and specific attributes?

Ans: Global attributes (like `class`, `id`) can be used on any element, while specific attributes apply to certain elements only (like `href` for `<a>`).

Q7. What does the `id` attribute do?

Ans: It uniquely identifies an element and is often used for CSS styling or JavaScript targeting.

Q8. What is the `data-*` attribute?

Ans: It allows storing custom data on HTML elements that can be accessed using JavaScript.

3. HTML Elements

Q9. What is an HTML element?

Ans: An HTML element consists of an opening tag, content, and a closing tag (e.g., `<p>Text</p>`).

Q10. What are void (self-closing) elements?

Ans: Elements that do not have closing tags, such as `<img>`, `<br>`, `<hr>`, and `<input>`.

Q11. What is the difference between block and inline elements?

Ans: Block elements take full width (e.g., `<div>`), while inline elements take only necessary width (e.g., `<span>`).

Q12. Can HTML elements be nested?

Ans: Yes, elements can be placed inside other elements as long as they are properly nested.

4. Forms and Inputs (4 Q&A)

Q13. What is the purpose of the `<form>` element?

Ans: It is used to collect user input and send it to a server for processing.

Q14. What are common input types in HTML?

Ans: `text`, `password`, `email`, `number`, `checkbox`, `radio`, `submit`.

Q15. What does the `name` attribute do in form inputs?

Ans: It identifies form data when the form is submitted.

Q16. What is the difference between the `GET` and `POST` methods?

Ans: `GET` sends data via URL, while `POST` sends data in the request body and is more secure.

Part 2: CSS

1. Selectors

Q1. What are CSS selectors?

Ans: CSS selectors are patterns used to target and style specific HTML elements.

Q2. What is the difference between class and id selectors?

Ans: Class selectors (`.class`) can be used multiple times, while ID selectors (`#id`) must be unique.

Q3. What are pseudo-class selectors?

Ans: They define a special state of an element, such as `:hover`, `:focus`, and `:nth-child()`.

2. Box Model

Q4. What is the CSS box model?

Ans: It consists of content, padding, border, and margin, which define the size and spacing of elements.

Q5. What does `box-sizing: border-box` do?

Ans: It includes padding and border inside the element's width and height.

Q6. What is the difference between padding and margin?

Ans: Padding adds space inside an element, while margin adds space outside it.

3. Positioning and Layout (3 Q&A)

Q7. What are the different CSS position values?

Ans: `static`, `relative`, `absolute`, `fixed`, and `sticky`.

Q8. What is the difference between Flexbox and Grid?

Ans: Flexbox is one-dimensional (row or column), while Grid is two-dimensional (rows and columns).

Q9. What does `z-index` do?

Ans: It controls the vertical stacking order of positioned elements.

4. Responsive Design (3 Q&A)

Q10. What is responsive web design?

Ans: It ensures web pages adapt to different screen sizes and devices.

Q11. What are media queries?

Ans: CSS rules that apply styles based on device characteristics like screen width.

Q12. What units are commonly used for responsive design?

Ans: `%`, `vw`, `vh`, `em`, and `rem`.

5. Styling (3 Q&A)

Q13. What is CSS specificity?

Ans: It determines which CSS rule is applied when multiple rules target the same element.

Q14. What is the difference between inline, internal, and external CSS?

Ans: Inline is written inside HTML tags, internal inside `<style>`, and external in a separate `.css` file.

Q15. What are CSS variables?

Ans: Custom properties (`--variable`) that allow reusable and maintainable styling values.

Part 3: JavaScript

1. DOM Manipulation (5 Q&A)

Q1. What is the DOM in JavaScript?

Ans: The DOM (Document Object Model) represents the HTML document as a tree structure that JavaScript can access and modify.

Q2. How do you select an element by ID in JavaScript?

Ans: Using `document.getElementById("id")`.

Q3. What is the difference between `querySelector` and `querySelectorAll`?

Ans: `querySelector` returns the first matching element, while `querySelectorAll` returns a NodeList of all matching elements.

Q4. How do you change the content of an HTML element?

Ans: By using `innerText`, `textContent`, or `innerHTML`.

Q5. How do you add an event listener to an element?

Ans: Using `addEventListener("event", callback)`.

2. Control Flow (5 Q&A)

Q6. What is control flow in JavaScript?

Ans: It determines the order in which statements are executed in a program.

Q7. What is the difference between `if...else` and `switch`?

Ans: `if...else` is best for complex conditions, while `switch` is better for multiple fixed values.

Q8. What are loops in JavaScript?

Ans: Loops repeat code execution; common types are `for`, `while`, and `do...while`.

Q9. What is the `break` statement used for?

Ans: It exits a loop or `switch` statement immediately.

Q10. What is the `continue` statement?

Ans: It skips the current iteration of a loop and continues with the next iteration.

3. ES6 Features (8 Q&A)

Q11. What is ES6?

Ans: ES6 (ECMAScript 2015) introduced modern JavaScript features for cleaner and more efficient code.

Q12. What are **let** and **const**?

Ans: Block-scoped variables; **let** allows reassignment, **const** does not.

Q13. What are arrow functions?

Ans: A shorter function syntax that does not bind its own **this**.

Q14. What is destructuring in JavaScript?

Ans: Extracting values from arrays or objects into variables.

Q15. What are template literals?

Ans: Strings defined using backticks that support interpolation and multiline text.

Q16. What is the spread operator?

Ans: **...** is used to expand arrays or objects.

Q17. What are default parameters?

Ans: Function parameters that have default values if none are provided.

Q18. What are modules in ES6?

Ans: They allow code to be split into reusable files using **import** and **export**.

4. APIs (7 Q&A)

Q19. What is an API in JavaScript?

Ans: An API allows JavaScript to interact with external systems or browser features.

Q20. What is the Fetch API?

Ans: A modern way to make HTTP requests using `fetch()`.

Q21. What is JSON?

Ans: A lightweight data format used for exchanging data between client and server.

Q22. What is the difference between synchronous and asynchronous code?

Ans: Synchronous code blocks execution, while asynchronous code runs without blocking.

Q23. What are Promises?

Ans: Objects representing the eventual completion or failure of an asynchronous operation.

Q24. What is `async/await`?

Ans: Syntax built on Promises to handle asynchronous code more cleanly.

Q25. What is Local Storage?

Ans: A Web API that stores data in the browser with no expiration date.